LIMITS OF DIVIDED POWER RINGS EXIST AND ARE COMPUTED ON RINGS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Limits from products and equalizers. categories-lemma-limits-products-equalizers (cross-chapter, named). Lifts the explicit product and equalizer constructions to all limits.

Source reference. Original reference 002N: focused-seeds/geometry/S9-dpa-limits/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The category of divided power rings has all limits and they agree with limits in the category of rings.*

Proof. The empty limit is the zero ring (that's weird but we need it). The product of a collection of divided power rings (A_t, I_t, γ_t), $t \in T$ is given by $(\prod A_t, \prod I_t, \gamma)$ where $\gamma_n((x_t)) = (\gamma_{t,n}(x_t))$. The equalizer of $\alpha, \beta : (A, I, \gamma) \to (B, J, \delta)$ is just $C = \{a \in A \mid \alpha(a) = \beta(a)\}$ with ideal $C \cap I$ and induced divided powers. It follows that all limits exist, see Categories, Lemma 002N. □

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.